

LLMsVerifier_API_SURFACE_QUICKREF

LLMsVerifier - API Surface Quick Reference

Package Hierarchy

```
digital.vasic.llmsverifier/  
├── llmverifier/      # Core verification engine  
├── providers/        # 25+ LLM provider adapters  
├── verification/     # Verification pipeline  
├── scoring/          # Scoring & ranking system  
├── api/              # REST API server  
├── database/         # SQLite + SQLCipher storage  
├── config/           # Configuration management  
├── client/           # HTTP client for LLM APIs  
├── auth/             # JWT + RBAC + LDAP auth  
├── capabilities/     # Dynamic capability detection  
├── ai/               # AI-powered recommendations  
├── analytics/        # Predictive analytics  
├── enhanced/         # Enterprise features  
├── tui/              # Terminal UI  
├── web/              # Web application  
├── sdk/              # Language SDKs  
├── messaging/        # Message broker factory  
├── monitoring/       # Metrics & monitoring  
├── scheduler/        # Job scheduler  
├── notifications/    # Notification system  
├── security/         # Security features  
├── failover/         # Failover mechanisms  
├── performance/     # Performance testing  
├── multimodal/       # Multimodal support  
├── events/           # Event system  
└── logging/         # Logging system
```

Top 50 Most Important Functions

Verification

Function	Package	Description
<code>llmverifier.New(cfg)</code>	<code>llmverifier</code>	Create Verifier instance
<code>llmverifier.LoadConfig(path)</code>	<code>llmverifier</code>	Load config from YAML
<code>Verifier.Verify()</code>	<code>llmverifier</code>	Run full verification
<code>Verifier.GenerateMarkdownReport(results, dir)</code>	<code>llmverifier</code>	Generate MD report
<code>Verifier.GenerateJSONReport(results, dir)</code>	<code>llmverifier</code>	Generate JSON report
<code>verification.NewVerifier(db)</code>	<code>verification</code>	Create verification service
<code>Verifier.Verify(ctx, req)</code>	<code>verification</code>	Verify single model
<code>NewCodeVerificationService(httpClient, logger)</code>	<code>verification</code>	Code verification service
<code>VerifyModelCodeVisibility(ctx, modelID, providerID, client)</code>	<code>verification</code>	"Do you see my code?" test
<code>NewCodingCapabilityVerificationService(httpClient, logger)</code>	<code>verification</code>	Coding capability service
<code>VerifyModelCodingCapabilities(ctx, modelID, providerID, client)</code>	<code>verification</code>	Full coding test

Providers

Function	Package	Description
<code>providers.NewProviderRegistry()</code>	<code>providers</code>	Create provider registry
<code>ProviderRegistry.GetConfig(name)</code>	<code>providers</code>	Get provider config
<code>NewModelProviderService(configPath, logger)</code>	<code>providers</code>	

Function	Package	Description
		3-tier discovery service
ModelProviderService.RegisterProvider(id, baseUrl, apiKey)	providers	Register provider
ModelProviderService.GetAllModelsWithVerification(ctx)	providers	Get all verified models
NewVerifiedConfigGenerator(service, logger, outputDir)	providers	Generate verified configs
VerifiedConfigGenerator.GenerateVerifiedConfig()	providers	Create verified config
NewErrorClassifier(provider)	providers	Classify API errors

Scoring

Function	Package	Description
scoring.NewScoringEngine(db, client, logger)	scoring	Create scoring engine
ScoringEngine.CalculateComprehensiveScore(ctx, modelID, config)	scoring	Calculate full score
scoring.DefaultScoringConfig()	scoring	Default scoring weights

API

Function	Package	Description
api.NewServer(cfg, db)	api	Create API server
Server.Start(port)	api	Start HTTP server
Server.Stop()	api	Stop server

Database

Function	Package	Description
database.New(path, encryptionKey)	database	Open encrypted SQLite
Database.CreateProvider(p)	database	Create provider
Database.GetProvider(id)	database	Get provider by ID
Database.ListProviders(filters)	database	List providers

Function	Package	Description
Database.CreateModel(m)	database	Create model
Database.GetModel(id)	database	Get model by ID
Database.ListModels(filters)	database	List models with filters
Database.CreateVerificationResult(r)	database	Save verification
Database.Ping()	database	Health check

Config

Function	Package	Description
config.LoadFromFile(path)	config	Load config (YAML/JSON/TOML)
config.SaveToFile(cfg, path)	config	Save config to file

Auth

Function	Package	Description
auth.NewAuthManager(jwtSecret, ldapConfig)	auth	Create auth manager
AuthManager.RegisterClient(name, desc, perms, rateLimit)	auth	Register API client
AuthManager.GenerateJWT(client)	auth	Generate JWT token
AuthManager.ValidateJWT(token)	auth	Validate JWT

Client

Function	Package	Description
client.New(baseUrl)	client	Create API client
Client.Login(username, password)	client	Authenticate
Client.GetModels()	client	List models via API

Capabilities

Function	Package	Description
capabilities.NewDetector()	capabilities	Create capability detector
Detector.DetectProviderCapabilities(ctx, provider, apiKey)	capabilities	Detect capabilities

Configuration Schema

Minimal Config

```
global:
  api_key: "${API_KEY}"
database:
  path: "./verifier.db"
api:
  port: "8080"
  jwt_secret: "secret"
```

Full Config

See main analysis document Section H.

API Endpoints

```
GET /api/health -> {"status": "healthy",
"timestamp": 1234}
GET /api/models?
provider_id=1&status=verified&min_score=7.0&search=gpt&limit=50
GET /api/models/{id}
POST /api/models/{id}/verify
GET /api/providers
POST /api/providers
```

Provider Support Matrix

Provider	File	Streaming	Vision	Function Calling	Code	Embeddings
OpenAI	openai.go	SSE	Yes	Yes	Yes	Yes
Anthropic	anthropic.go	SSE	Yes	Yes	Yes	Yes
Groq	groq.go	SSE	Yes	Yes	Yes	No
DeepSeek	deepseek.go	SSE	No	Yes	Yes	Yes
Cohere	cohere.go	SSE	No	Yes	Yes	Yes
Mistral	mistral.go	SSE	No	Yes	Yes	Yes
xAI	xai.go	SSE	No	Yes	Yes	No
Replicate	replicate.go	No	Yes	No	Yes	No
Cerebras	cerebras.go	SSE	No	Yes	Yes	No
Cloudflare	cloudflare.go	SSE	No	No	Yes	Yes
SiliconFlow	siliconflow.go	SSE	Yes	Yes	Yes	Yes
Together AI	togetherai.go	SSE	Yes	Yes	Yes	Yes
Hyperbolic	hyperbolic.go	SSE	No	Yes	Yes	Yes

Provider	File	Streaming	Vision	Function Calling	Code	Embeddings
SambaNova	sambanova.go	SSE	No	Yes	Yes	No
Moonshot/ Kimi	kimi.go	SSE	No	Yes	Yes	No
Qwen/ DashScope	qwen.go	SSE	Yes	Yes	Yes	Yes
10+ more	...	...	...	...	...	...

Scoring Weights

```
DefaultScoreWeights() = {
    ResponseSpeed:    0.25 // Latency
    ModelEfficiency:  0.20 // Throughput, context window
    CostEffectiveness: 0.25 // Price per token
    Capability:       0.20 // Feature breadth
    Recency:          0.10 // Last update date
}
```

Score range: 0.0 - 10.0

Database Schema (Key Tables)

```
providers          -- Provider configurations
models             -- Model metadata and capabilities
verification_results -- Verification outcomes
api_keys           -- API key storage
config_exports     -- Exported configurations
events            -- System events
issues             -- Detected issues
schedules          -- Verification schedules
pricing            -- Model pricing data
users              -- User accounts
sessions           -- Active sessions
migrations         -- Schema migrations
verification_scores -- Calculated scores
```

Integration for HelixTranslate

```
// 1. Import
go get digital.vasic.llmsverifier

// 2. Load config
cfg, _ := config.LoadFromFile("config.yaml")

// 3. Create verifier
v := llmverifier.New(cfg)
```

```
// 4. Verify
results, _ := v.Verify()

// 5. Score
db, _ := database.New(cfg.Database.Path, "")
engine := scoring.NewScoringEngine(db, modelsDevClient, logger)
score, _ := engine.CalculateComprehensiveScore(ctx, "model-id",
    scoring.DefaultScoringConfig())

// 6. Report
v.GenerateMarkdownReport(results, "./reports")
```